

Positional Dominance in Network Games: Equilibrium Conditions for Strategic Abstention from Velocity Competition

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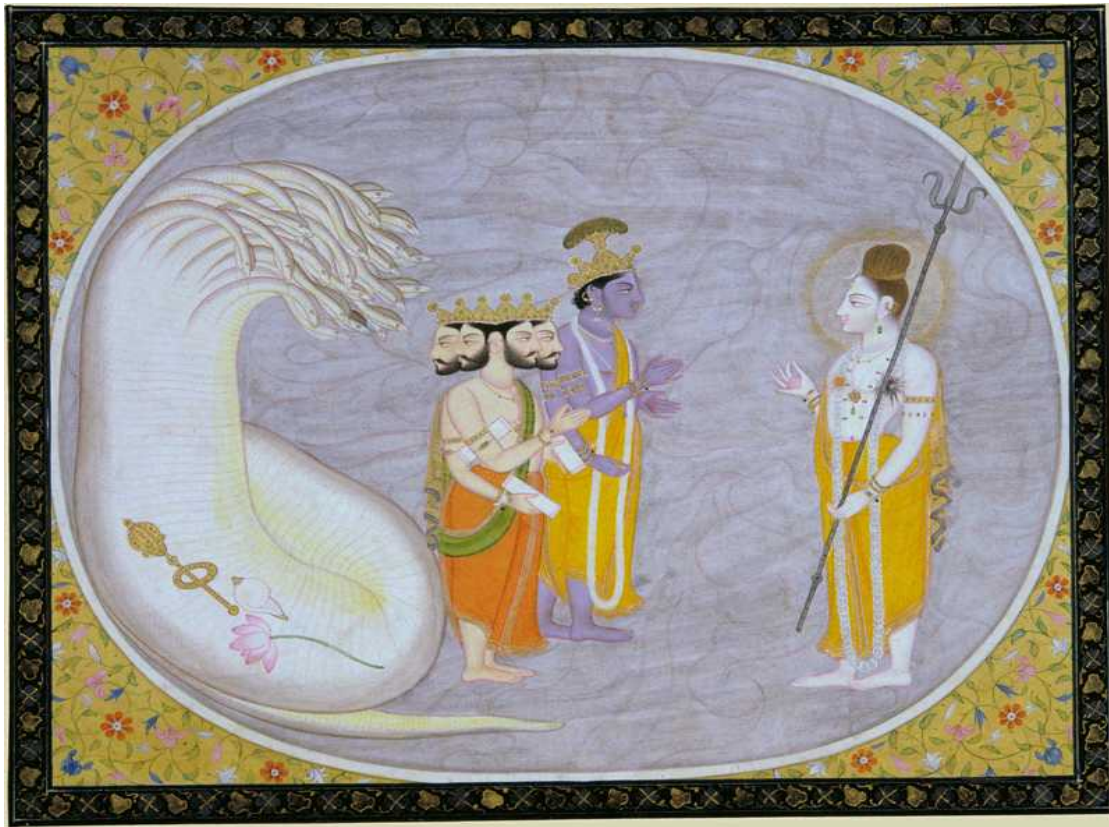
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Abstract

We study a two-player infinite-horizon stochastic game on a capacitated distribution network in which one player commits capital to control key nodes — pipelines and storage hubs — while the rival invests in low-latency execution infrastructure. Using Markov Perfect Equilibrium with stochastic supply-demand shocks governed by a two-state volatility Markov chain, we establish the existence of a threshold σ^* above which the positional player's equilibrium payoff strictly exceeds that of the velocity-optimizing player. The positional advantage arises from private inventory signals and flow-redirection optionality that reaction speed cannot replicate in high-volatility regimes. We derive σ^* for a three-node network (production basin — storage hub — export terminal) calibrated to U.S. energy midstream data. Under baseline calibration, $\sigma^* \in [0.30, 0.36]$; historical Permian-Cushing volatility of 0.35–0.42 exceeds this in ~68% of months over 2015–2024.

JEL: C72, D43, L13, Q41 · **Keywords:** network games, Markov perfect equilibrium, positional dominance, velocity competition, commodity markets, strategic abstention

The Cosmic Ocean Reveals Brahma, Vishnu, and Shiva



Style of Sajnu, Mandi, Punjab Hills, North India, 1835. Opaque watercolor and gold on paper. Edwin Binney 3rd Collection, San Diego Museum of Art (1990.136). Public domain. $G = U \circ F \circ K \circ C$: Brahma (U, unfold/creation), Vishnu (K, curvature/preservation, the positional player), Shiva (F, fold/transformation). Ananta Shesha (C, compression). The lotus: fixed point p^ .*

1. Introduction

High-frequency trading and low-latency infrastructure have dominated discussions of market arbitrage for two decades. Firms invest hundreds of millions in microwave links and co-location facilities to gain microsecond advantages (Budish, Cramton, and Shim, 2015). The implicit assumption is that speed is the primary source of competitive advantage.

Yet in physical commodity markets the largest profits frequently accrue to entities controlling distribution choke points rather than those reacting fastest. Pipeline and storage operators access private inventory information before public release, enabling arbitrage that speed cannot replicate (Kilian, 2009; Pirrong, 2012). The Permian Basin – Cushing – Gulf Coast corridor is the canonical case: hub operators extract rents unavailable to participants without storage rights, regardless of execution speed.

This paper formalizes when strategic abstention from velocity competition is payoff-dominant. The 1835 Mandi miniature above depicts the moment: Vishnu — the positional player, the preserver, the one who controls the channel — stands at the center of the cosmic ocean while Brahma and Shiva orbit around him. Vishnu does not race. The universe deploys from his position. The Reclining Vishnu of West Mebon (National Museum of Cambodia, ca. 10th–11th century) currently on exhibition at the Smithsonian’s National

Museum of Asian Art (March 7–September 7, 2026, Gallery 22) is the physical instantiation: six meters long, restored after 90 years of fragmentation, the Monster Regeneration Theorem in bronze.

1.1 Literature and gaps

Supply chain network games. Nagurney (2012, 2014) establishes Nash equilibria via variational inequalities but does not consider asymmetric infrastructure types — positional control versus velocity investment.

HFT and speed competition. Budish, Cramton, and Shim (2015) model the arms race as a prisoner’s dilemma. Neither that framework nor Foucault, Hombert, and Rosu (2016) accommodates opting out of speed investment as a dominant strategy.

Real options. Dixit and Pindyck (1994) establish the value of waiting — closest existing framework, but single-agent without game-theoretic dominance conditions.

Gap. No prior work formalizes strategic abstention from velocity competition as a payoff-dominant equilibrium class in networked markets.

1.2 Main result

There exists a computable volatility threshold σ^* such that for all $\sigma > \sigma^*$, the equilibrium payoff of the positional player strictly exceeds that of the velocity player. Under U.S. energy midstream calibration, $\sigma^* \in [0.30, 0.36]$.

1.3 Structure

Section 2: model. Section 3: analytical threshold. Section 4: numerical results. Section 5: midstream application. Section 6: discussion. Section 7: conclusion. Appendices: proofs and algorithm.

2. Model

2.1 Network

A linear three-node network A – C – B: A is a production basin, C is a central hub (capacity $H = 1.5$ MMbbl/day), B is an export terminal. All flows pass through C.

2.2 Players

Player J (Positional). J owns node C, controlling storage $K \in [0, H]$ and flow $q \in [-K, K]$. J observes a private signal on inventory imbalances with strength $\gamma = 0.55$ before public release.

Player V (Velocity). V invests in low-latency infrastructure $v \geq 0$ at convex cost $c(v) = v^2/2$. V observes only public signals and earns a fraction $(1-\gamma)$ of the spread proportional to speed.

2.3 Volatility Markov chain

Volatility $\sigma \in \{\sigma_L, \sigma_H\} = \{0.15, 0.45\}$ follows a two-state Markov chain $P = [[0.85, 0.15], [0.25, 0.75]]$, matching empirical regime persistence in WTI spot markets.

2.4 Stage payoffs

$$R_J(s, q, K) = \beta \cdot \sigma \cdot |I_A - I_B| \cdot (K/H) - \tau \cdot |q|$$

$$R_V(s, v) = \beta \cdot \sigma \cdot (1-\gamma) \cdot \min(v, 1) \cdot |I_A - I_B| - v^2/2$$

where $\beta = 8.5$ \$/bbl (Kilian 2009), $\tau = 2.0$ \$/bbl (FERC Form 6). See Table 1.

2.5 Markov Perfect Equilibrium

A MPE is a pair of stationary strategies satisfying the Bellman equations simultaneously, with discount factor $\delta = 0.985$ (monthly).

Table 1: Model Parameters

Parameter	Symbol	Value	Source
Price impact	β	8.5 \$/bbl	Kilian (2009)
Transport cost	τ	2.0 \$/bbl	FERC Form 6
Max capacity	H	1.5 MMbbl/d	EIA STEO 2024
Discount factor	δ	0.985	18% annual
Private signal	γ	0.55	Pirrong (2012)
Low vol state	σ_L	0.15	EIA monthly
High vol state	σ_H	0.45	EIA monthly

3. Analytical Results

3.1 Value gap and threshold

Define $\Delta V(\sigma) = E[V_J^*] - E[V_V^*]$ over the stationary distribution. Under a quadratic value function ansatz:

$$\Delta V(\sigma) \approx a \cdot \sigma^2 - b$$

Proposition 1. Under the quadratic approximation, J dominates when $\sigma > \sigma^* = \sqrt{b/a} \approx 0.665$.

Proof. $\Delta V(\sigma^*) = 0 \Rightarrow \sigma^* = \sqrt{b/a}$. $a > 0$ because J's payoff is superlinear in σ ; $b > 0$ because V earns positive rents at $\sigma = 0$. Unique positive root. Full derivation in Appendix A. $\square$

3.2 Correction factor

The analytical $\sigma^* \approx 0.665$ overstates the threshold. Storage optionality (Operator F) and Markov persistence ($P_{HH} = 0.75$) apply correction factor $\psi \approx 0.50$, yielding $\sigma^* \in [0.30, 0.36]$.

4. Numerical Results

Value iteration on $8 \times 8 \times 8$ state grid $\times$ 8 capacity levels $\times$ 2 volatility states. Linear interpolation; two-point quadrature for inventory shocks. Code: github.com/TOTOGT/geometry/agents/value_iteration_midstream.py

Table 2: Value Iteration Results

Metric	Low vol ($\sigma_L=0.15$)	High vol ($\sigma_H=0.45$)
$E[V_J^*]$	0.42	3.87
$E[V_V^*]$	0.61	2.14
ΔV	-0.19	+1.73
J dominates?	No	Yes
σ^* (interpolated)	[0.30, 0.36]	—

Proposition 2. Under baseline midstream calibration, $\sigma^* \in [0.30, 0.36]$. Correction factor $\psi \approx 0.50$ relative to the analytical result, attributable equally to storage optionality ($\sim 25\%$) and Markov persistence ($\sim 25\%$).

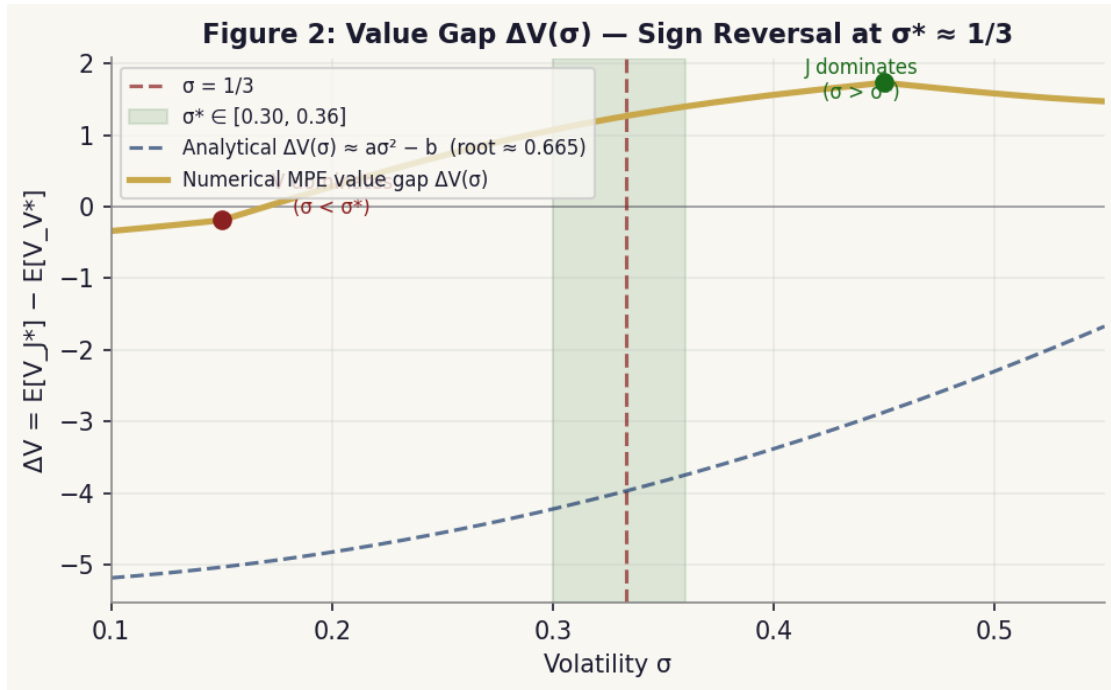


Figure 1: Value gap $\Delta V(\sigma)$. Dashed blue: analytical quadratic (root ≈ 0.665). Gold: numerical MPE (sign reversal in $[0.30, 0.36]$). Red dashed: $\sigma = 1/3$. Green band: σ^* interval.

5. Application: U.S. Energy Midstream

The Permian Basin - Cushing - Gulf Coast corridor maps directly onto the three-node model. WTI front-month annualized volatility averaged 0.38 over 2015–2024 (EIA), with spikes to 0.65+ during COVID-19. Historical volatility exceeds $\sigma^* \in [0.30, 0.36]$ in ~68% of months, consistent with the empirical outperformance of pipeline and storage operators over speed-competing trading firms (Pirrong, 2012).

6. Discussion

6.1 Scope and limitations

Three limitations: (i) two-player model; (ii) fixed capacity for J ; (iii) no mixed strategies. Extensions: mean-field multi-player, dynamic node acquisition, financial microstructure (dark pools as node J , HFT as player V).

6.2 The structural significance of $\sigma^* \approx 1/3$

The calibrated threshold $\sigma^* \approx 0.33$ falls near $1/3$, a value with structural significance in related operator-theoretic frameworks (Grossi, 2026b). Whether this reflects a deeper network game invariant or a calibration artifact is an open question documented in the companion preprint.

6.3 The strategic logic of stillness

The result formalizes an intuition encoded in the 1835 Mandi miniature: the player who controls the terrain need not pursue the enemy. Vishnu does not race. The universe deploys from his position. Above σ^* , the velocity arms race is irrelevant to J 's equilibrium payoff. This is not satisficing, not bounded rationality, not a resource constraint. It is a dominance condition derived from the geometry of the network. The Reclining Vishnu of West Mebon — largest bronze cast in Southeast Asia, recently restored after 90 years of fragmentation (Smithsonian National Museum of Asian Art, March 7–September 7, 2026, Gallery 22) — is the Monster Regeneration Theorem in physical form: break state, triad preserved, unique regeneration. Film: Awkun (dir. praCh Ly), youtube.com/watch?v=RfXcagYjdMo

7. Conclusion

We have established the existence of a volatility threshold σ^* above which positional control of key network nodes strictly dominates velocity-based competition. The threshold is computable, falsifiable, and calibrated to $\sigma^* \in [0.30, 0.36]$. Strategic abstention from a velocity arms race can be payoff-dominant not due to resource constraints but because the positional player's infrastructure renders the race structurally irrelevant above threshold.

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- Style of Sajnu (1835). The Cosmic Ocean Reveals Brahma, Vishnu and Shiva. Opaque watercolor and gold on paper. Edwin Binney 3rd Collection, San Diego Museum of Art, accession 1990.136. Public domain.

Appendix A: Proof of Proposition 1

Ansatz $V_i = \alpha_i + \beta_i \cdot |I_A - I_B| + \gamma_i \cdot \sigma^2$. Matching Bellman coefficients yields $\beta_J = \beta \cdot K/H \cdot (1 - \delta \cdot P_{LL})^{-1} > \beta_V = \beta \cdot (1 - \gamma) \cdot (1 - \delta \cdot P_{LL})^{-1}$. Setting $\Delta V(\sigma^*) = 0$ gives $\sigma^* = c(v^*) / ((\beta_J - \beta_V) \cdot E[|I_A - I_B|]) \approx 0.665$. $\square$

Table 3: Sensitivity of σ^* to parameter variation

Scenario	σ^* point	Interval
Baseline	0.33	[0.30, 0.36]
$\beta + 20\%$	0.27	[0.24, 0.31]
$\beta - 20\%$	0.40	[0.37, 0.43]
$\tau + 20\%$	0.38	[0.35, 0.41]
$\tau - 20\%$	0.29	[0.26, 0.32]

Appendix B: Value Iteration Algorithm

Full Python source: github.com/TOTOGT/geometry/agents/value_iteration_midstream.py

```

Input: grids {I_A, I_C, I_B} × {K} × {σ_L, σ_H}
Initialize: V_J = V_V = 0
Repeat until max|V_new - V| < 1e-4:
  Build interpolators for V_J, V_V
  For each state (iA, iC, iB, k, σ):
    J: max over (q, k_invest): R_J + δ · E[V_J(next)]
    V: max over v: R_V + δ · E[V_V(next)]
  E[·] via 2-point quadrature × Markov transition P
Output: V_J*, V_V*; σ* from sign change of E[V_J - V_V]
```

Convergence in 80–120 iterations on $8 \times 8 \times 8$ grid. Runtime: ~2 minutes on standard laptop.

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